

AI Acceptance in the Sharing Economy

Project progress for coaching session

May 28, 2026

1 Research question

How do users perceive AI agents on peer-to-peer rental platforms, and what drives trust — across platform types and across community discussion (Reddit) vs. paying-customer experience (Trustpilot)?

The sharing economy increasingly embeds AI agents that automate matching, messaging, pricing and damage assessment. We study how users *react* to that automation and which design factors earn (or lose) their trust.

2 Data we collected ourselves

Table 1: Two self-scraped sources. Trustpilot star ratings give a ground-truth sentiment label; Reddit is labelled by community theme.

Source	Documents	Groups	From	To
Reddit (balanced)	483	2	2021-11-17	2026-05-26
Trustpilot reviews	3912	7	2019-07-04	2026-05-28

Reddit was scraped with `RedditExtractor` targeting sharing-economy and AI subreddits; Trustpilot with an Apify actor across seven rental platforms. Each captures a different lens on the same question.

3 What users talk about (word frequency)

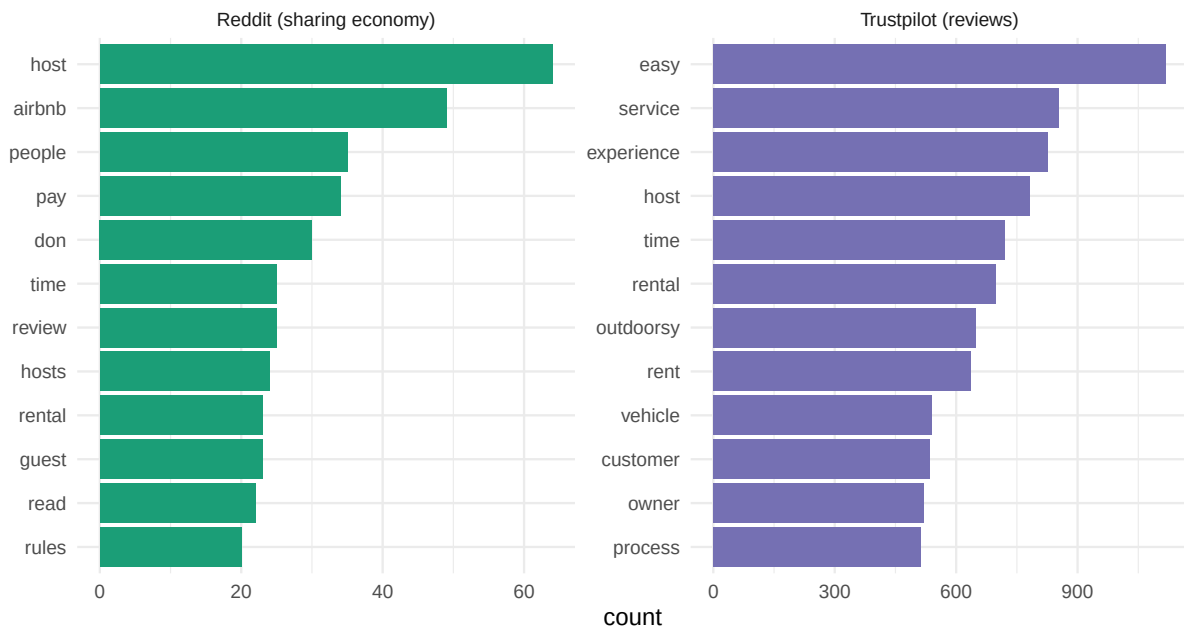


Figure 1: Top terms differ sharply by source: Reddit centres on the host–renter relationship and disputes; Trustpilot on transactional service quality.

4 Trust varies by platform type (Trustpilot ratings)

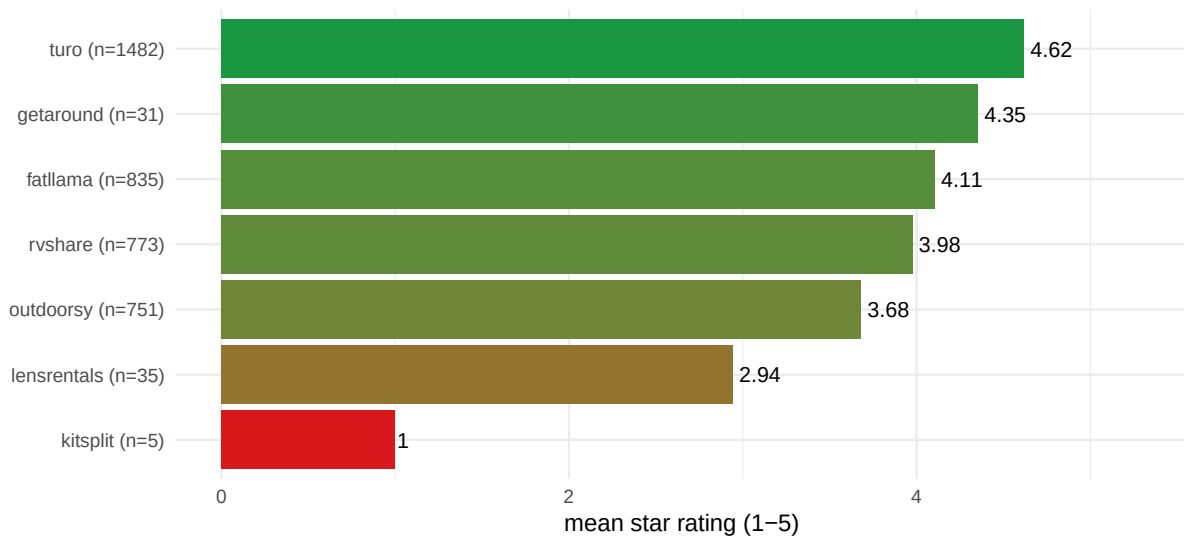


Figure 2: Mean star rating by platform (n shown). Peer-to-peer mobility (Turo, RVshare, Outdoorsy, Fat Llama) is trusted; specialist equipment rental scores far lower – though those samples are small.

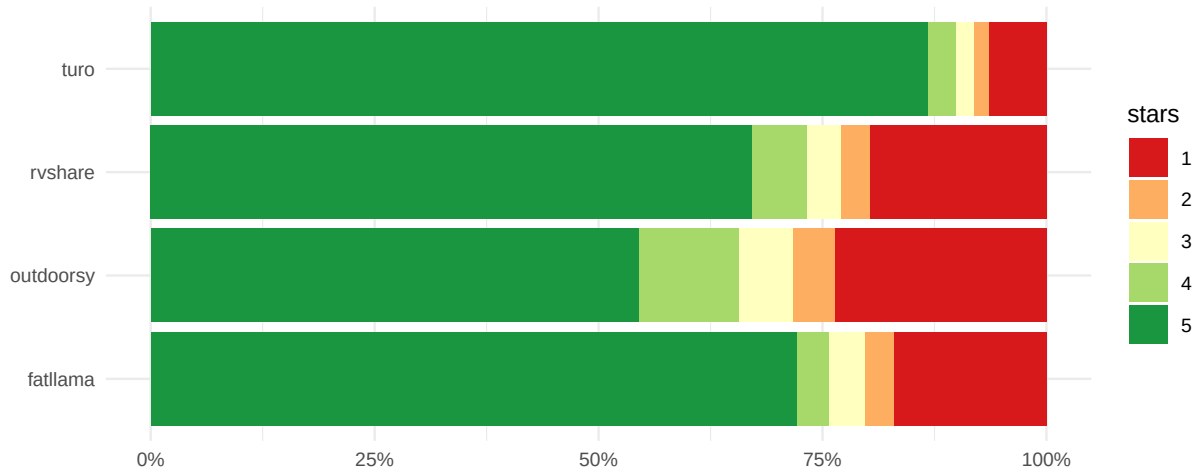


Figure 3: Rating mix per platform (platforms with ≥ 50 reviews). The positive skew is typical of Trustpilot; we will downsample for balanced classification.

5 Sentiment over time

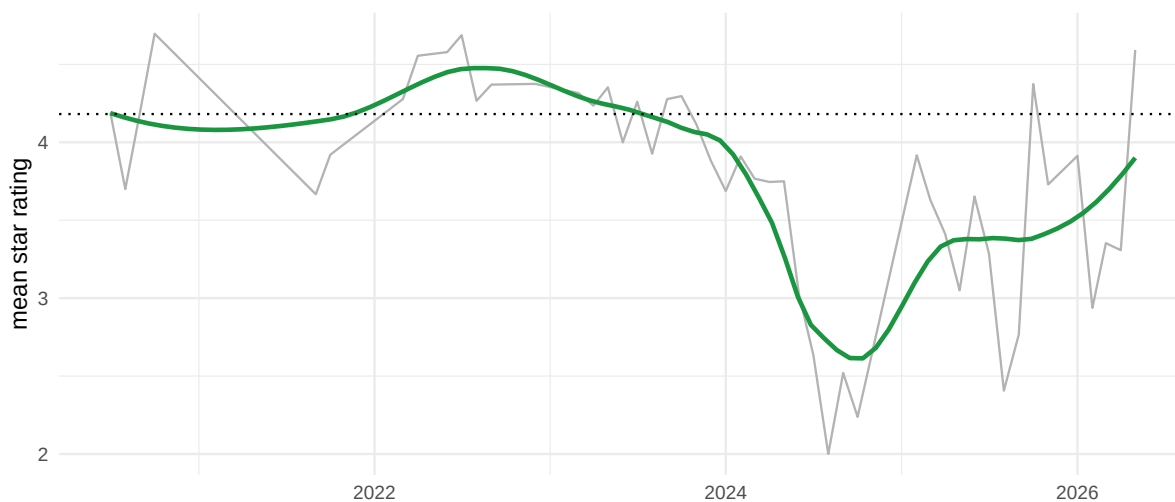


Figure 4: Monthly mean Trustpilot rating, 2019-2026 (months with ≥ 15 reviews). Satisfaction is broadly stable with a mild dip in the post-2022 travel surge; this is the timeline we will extend with lexicon sentiment on the Reddit text.

6 Distinctive vocabulary (comparison cloud)



Figure 5: Comparison cloud: words coloured by the source where they are relatively more frequent. Reddit (green) skews to disputes and decisions; Trustpilot (purple) to service and ease.

7 Where we are & next steps

Done: two on-topic corpora scraped and cleaned (Reddit 483 comments, Trustpilot 3912 reviews across 7 platforms); exploratory word frequency, ratings, timeline and comparison cloud above.

Next (planned):

- **Preprocessing:** lemmatise, 1-gram + bi-gram tokens.
- **Topic modelling (LDA)** 3–5 topics per class, seeded with trust / price / service / AI terms; compare with `text2vec`.
- **Sentiment:** AFINN/NRC on Reddit + validate against Trustpilot stars; 8-emotion (NRC)

profile per source.

- **Embeddings** (GloVe) for semantic neighbours of *trust*, *AI*, *host*.
- **Synthesis**: map topics to trust frameworks (Mayer 1995; TAM) and produce actionable **AI-agent design recommendations**.

One decision for the professor: keep our *cross-source / cross-platform comparison* framing, or pivot to a stricter *positive-vs-negative classification*? This shapes the LDA and labelling setup.

Known limitations: Trustpilot skews positive; equipment-rental platforms under-sampled; “trust” appears mostly implicitly; English-only.